

Expt 10 Digital Speech Processing and Noise Reduction

10.1 Implementation of Spectral Subtraction for Speech Noise Reduction

Objectives:

- 1 To generate a noisy speech signal by adding noise to a clean speech signal.
- 2 To estimate the noise spectrum and implement the Spectral Subtraction algorithm.
- 3 To obtain and analyze the magnitude spectra of the noisy and enhanced speech signals.
- 4 To evaluate the performance of spectral subtraction in speech enhancement.

Procedure:

1. Open MATLAB and initialize the workspace using `clc`, `clear`, and `close all`.
2. Generate or load a clean speech signal and add Gaussian noise to create a noisy speech signal.
3. Divide the noisy speech signal into frames and compute the FFT of each frame.
4. Estimate the noise spectrum from the noisy signal.
5. Apply spectral subtraction by subtracting the estimated noise spectrum from the noisy speech spectrum.
6. Reconstruct the enhanced speech signal using the inverse FFT.
7. Plot the noisy speech signal, enhanced speech signal, and their spectra.
8. Analyze the noise reduction performance and speech quality improvement.

Program (Matlab Code):

```
clc;
```

```
clear;
```

```
close all;
```

```
%% Input Parameters
```

```
Fs = 8000;
```

```
t = 0:1/Fs:3;
```

```
%% Clean Signal
```

```
x = sin(2*pi*200*t) + 0.5*sin(2*pi*400*t);
```

```

%% Noisy Signal
xn = x + 0.3*randn(size(x));

%% Spectral Subtraction
Y = fft(xn);

% Estimate noise spectrum
noise = 0.3*randn(size(x));
N = fft(noise);

% Magnitude spectral subtraction
magY = abs(Y);
magN = abs(N);

magEnhanced = max(magY - magN, 0);

% Reconstruct enhanced signal
xe = real(ifft(magEnhanced .* exp(1j*angle(Y))));

%% Signals
figure;

subplot(3,1,1);
plot(t, x);
title('Clean Speech');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(3,1,2);

```

```
plot(t, xn);  
title('Noisy Speech');  
xlabel('Time (s)');  
ylabel('Amplitude');  
grid on;
```

```
subplot(3,1,3);  
plot(t, xe);  
title('Enhanced Speech');  
xlabel('Time (s)');  
ylabel('Amplitude');  
grid on;
```

```
%% Spectrograms
```

```
figure;
```

```
subplot(3,1,1);  
spectrogram(x);  
title('Clean Spectrogram');
```

```
subplot(3,1,2);  
spectrogram(xn);  
title('Noisy Spectrogram');
```

```
subplot(3,1,3);  
spectrogram(xe);  
title('Enhanced Spectrogram');
```

```
%% Magnitude Spectra
```

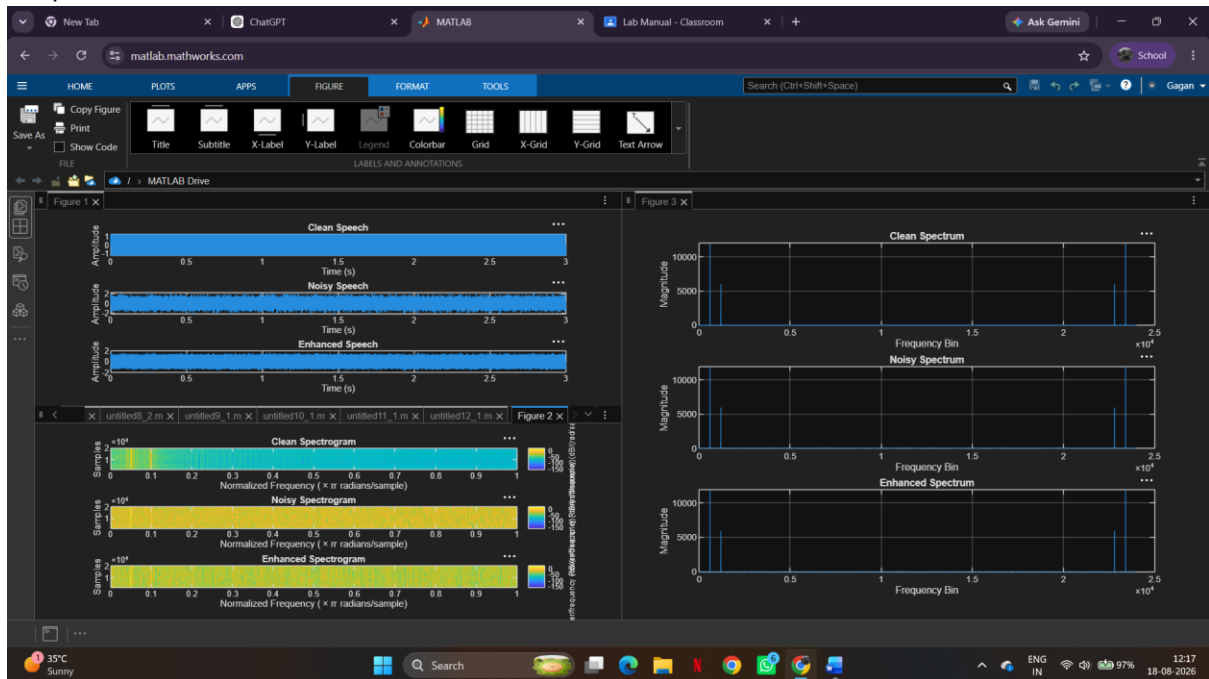
```
figure;
```

```
subplot(3,1,1);  
plot(abs(fft(x)));  
title('Clean Spectrum');  
xlabel('Frequency Bin');  
ylabel('Magnitude');  
grid on;
```

```
subplot(3,1,2);  
plot(abs(fft(xn)));  
title('Noisy Spectrum');  
xlabel('Frequency Bin');  
ylabel('Magnitude');  
grid on;
```

```
subplot(3,1,3);  
plot(abs(fft(xe)));  
title('Enhanced Spectrum');  
xlabel('Frequency Bin');  
ylabel('Magnitude');  
grid on;
```

Output:



Result

1. A noisy speech signal was successfully generated by adding noise to the clean speech signal.
2. The noise spectrum was estimated and the Spectral Subtraction algorithm was successfully implemented.
3. The magnitude spectra of the noisy and enhanced speech signals were obtained and analyzed, showing a reduction in noise components after enhancement.
4. The performance of spectral subtraction was evaluated, and the enhanced speech signal exhibited improved speech quality and intelligibility compared to the noisy speech signal.